

Donnell Naidoo

Johannesburg | 0681581985 | donnellnaidoo6@gmail.com | <https://donnellnaidoo.vercel.app/>

OBJECTIVE

Early-career software developer, AI practitioner, and educator with experience building full-stack applications, data-driven solutions, and mentoring university students. Passionate about using technology to solve real-world problems and improve outcomes for communities, organizations, and individuals. Combines strong technical skills with communication, teaching, and business analysis capabilities to bridge the gap between complex technology and practical impact.

EDUCATION

University of Johannesburg (2023-2025)

Diploma in Business Information Technology (Cum-Laude)

Graduated in 2026

- Modules: Software Development, Information Systems, Systems Software, Communication Networks, Business Analysis

University of Johannesburg (2026-2027)

Advanced Diploma in Business Information Technology

Graduate in 2027

- Modules: Advanced Programming, Advanced Database, Software Requirements (Software Engineering), Advanced Business Analysis

EXPERIENCE

TechXplo (FREELANCE)

June 2025 - PRESENT

Full Stack Software Developer

- Designed and implemented software solutions that improved operational efficiency and streamlined business processes for internal and client-facing systems.
- Design and implement responsive user interfaces and scalable backend services for internal systems and client solutions.
- Collaborated with multidisciplinary teams to translate business requirements into scalable technical solutions.
- Build and integrate RESTful APIs, databases, and authentication systems for secure application functionality.
- Developed dashboards and data visualizations that enabled stakeholders to make informed decisions using real-time operational insights.
- Use Jira and Confluence for agile project management, sprint tracking, documentation, and technical collaboration.
- Participate in debugging, testing, and optimization of Windows and Linux-based application environments.
- Assist with software documentation, technical writing, and maintenance of development standards and security best practices.

University of Johannesburg

February 2025 - PRESENT

Software Development Tutor

- Mentored 100+ first- and second-year software development students through programming and software engineering coursework.
- Translated complex technical concepts into practical explanations tailored to diverse learning styles.
- Facilitated collaborative problem-solving sessions that improved student confidence and technical competency.
- Guided students through debugging, project development, and software design decisions.
- Developed communication and instructional skills by presenting technical material to audiences with varying levels of technical knowledge.

SELECTED AI PROJECTS

AI-Powered Crime Detection Platform (In Development)

- Designing and developing an AI-powered video analysis platform that detects suspicious and criminal activities from surveillance footage using computer vision and deep learning models.
- Fine-tuning and evaluating transformer-based video classification models to identify incidents such as theft, vandalism, fighting, and other security-related events.
- Building a full-stack application that enables video ingestion, automated incident detection, event monitoring, and alert management through an intuitive user interface.
- Developing dashboards and reporting tools that help security personnel review incidents, analyze trends, and make informed operational decisions.
- Exploring scalable deployment strategies for real-world environments, including cloud-hosted AI inference, model optimization, and system monitoring.
- Applying responsible AI practices through performance evaluation, false-positive analysis, and continuous model improvement to enhance reliability and trustworthiness.

ONESA – AI-Powered Civic Transparency Platform

- Developed a civic technology platform that improves transparency between South African citizens and government stakeholders.
- Implemented sentiment analysis to evaluate citizen feedback on public services and tender projects.
- Built features that enable citizens to track project progress and share community feedback.
- Developed dashboards that transform public sentiment and engagement data into actionable insights.
- Applied AI and data analytics to identify trends, recurring concerns, and service delivery issues.
- Led end-to-end development, including system design, full-stack implementation, database integration, and AI features.

AI-Powered Insurance Coverage Assistant

- Designed and developed an AI-powered insurance assistant using the ChatGPT Assistants API to support insurance brokers in policy analysis and coverage determination.
- Built a document-processing workflow that allows brokers to upload policy wording and query policy documents using natural language.
- Implemented retrieval-based document analysis to help users identify relevant clauses, exclusions, conditions, and coverage provisions within complex insurance contracts.
- Developed functionality that assists brokers in evaluating claims scenarios against policy wording and generating evidence-based coverage assessments.
- Improved the speed and accessibility of policy interpretation by reducing the time required to manually review lengthy insurance documentation.
- Focused on transparency and responsible AI usage by ensuring responses referenced supporting policy language rather than relying solely on model-generated conclusions.
- Delivered an end-to-end solution including document ingestion, AI integration, backend services, and user-facing interfaces.

Academic & Learning Support Initiatives

- Leveraged AI tools and software development expertise to create learning resources and practical demonstrations for university students.
- Simplified complex software engineering and programming concepts through structured explanations and hands-on examples.
- Supported student problem-solving and technical development through mentoring, tutoring, and guided project assistance.
- Demonstrated the ability to communicate technical concepts effectively to audiences with varying levels of technical experience.

SKILLS & CERTIFICATIONS

SKILLS

Programming Languages

- Python • JavaScript • TypeScript • SQL • C#

Frontend & Mobile Development

- React • Next.js • React Native • HTML • CSS • Tailwind CSS

Backend & Databases

- .NET • REST APIs • PostgreSQL • SQL Server • Database Design

Artificial Intelligence & Data

- Large Language Models (LLMs) • Prompt Engineering • Sentiment Analysis • Machine Learning Fundamentals
• Data Analysis • Power BI • Tableau • Microsoft Fabric

Tools & Platforms

- Git • GitHub • Jira • Confluence • Figma • Microsoft Office Suite

Software Engineering

- Agile Development • Software Requirements Engineering • Business Analysis • System Design • API Integration
• Technical Documentation

Leadership & Communication

- Technical Mentoring • Public Speaking • Problem Solving • Critical Thinking • Team Collaboration • Stakeholder Communication